Double-Lined from the Archive: Independent Dynamical Masses of M Dwarfs in Eclipsing Binaries from Public HARPS and ESPRESSO Spectra

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Abstract

An eclipsing binary of a solar-type star and an M dwarf gives model-independent masses for both stars once the M dwarf's own lines are measured, at optical flux ratios near 10^{-3}. This paper measures them from public ESO archive spectra with an analysis built independently of the published ones. Each spectrum's velocity is measured against a template and the primary's orbit fitted to those velocities; the spectra are placed on one velocity grid in the primary's frame and divided by their series' median; the residuals are cross-correlated by inverse variance with an empirical M1 V spectrum; the star's pattern, the sky's pattern and five principal components are removed; and the rows are summed along the companion's track over its semi-amplitude K_2 and its systemic offset, against nulls with the orbit's phase moved and with whole nights shuffled. From 260 ESPRESSO spectra of TOI-1338/BEBOP-1, 114 HARPS spectra of EBLM J0310−31 and 35 HARPS spectra of HD 4875 the companions stand at 19.3, 19.2 and 13.3σ, with $K_2 = 77.27$, 88.12 and 75.29 $\mathrm{km\,s^{-1}}$ against published values of 77.16 ± 0.48, 88.27 ± 0.46 and 75.13 ± 0.16 $\mathrm{km\,s^{-1}}$, and masses within 0.5 per cent of the published dynamical masses; HD 4875's companion had been measured only in the near-infrared. In all three the companion's lines sit +1.05 to +1.25 $\mathrm{km\,s^{-1}}$ from the primary's systemic velocity, and holding that offset at zero moves K_2 by up to 0.55 $\mathrm{km\,s^{-1}}$. Across the 35 searchable EBLM binaries with archived HARPS series, six companions stand above 5σ, four of them without a published semi-amplitude: EBLM J2101−45 at 15.0σ, whose 75 spectra give $K_2 = 65.41$ $\mathrm{km\,s^{-1}}$ and minimum masses of 1.431 and 0.556 $\mathrm{M_\odot}$, and J0649−27, J0825−67 and J1141−37 at 6.7, 6.4 and 5.2σ.

1 Introduction

Masses and radii of M dwarfs set the properties inferred for the planets that orbit them, and the empirical relations behind those inferences rest on stars whose masses are measured without models. Detached double-lined eclipsing binaries provide such masses, but few are known below the fully convective boundary near 0.35 $\mathrm{M_\odot}$ [1, 2], since an M dwarf paired with a solar-type star contributes about a thousandth of the optical light and its lines are lost in the primary's. The EBLM project has characterised hundreds of these single-lined eclipsing binaries [3, 4, 5], measuring the primary's spectroscopic orbit and the eclipses and taking the primary's mass or radius from stellar models. The BEBOP survey for circumbinary planets has added years of precise velocities of the same stars with HARPS and SOPHIE [6, 7, 8].

The secondary's lines can nonetheless be recovered. High-resolution cross-correlation, developed to detect molecules in exoplanet atmospheres [9, 10], sums a weak spectrum along a known orbit after the stationary spectrum has been removed. Sebastian et al. applied it to the circumbinary planet host TOI-1338/BEBOP-1 [11, 12] and detected its M dwarf at 11σ in 103 ESPRESSO spectra, measuring $K_2 = 77.16 \pm 0.48$ $\mathrm{km\,s^{-1}}$ and masses of 1.098 ± 0.017 and 0.307 ± 0.003 $\mathrm{M_\odot}$ [13], with the Saltire model of the cross-correlation map [14]. The method has since been applied

to Kepler-16 with SOPHIE [15], to BEBOP-3 [16] and to five more EBLM binaries, among them EBLM J0310−31 [17]. In the near-infrared, where the M dwarf is relatively brighter, the secondary's semi-amplitude has been measured with SPIRou [18], CRIRES+ [19] and NIRPS [20, 21].

These measurements share a small number of analysis designs, and their accuracy has been tested by splitting each dataset in time and by simulation [13, 17]. An analysis built independently, with a different template, a different removal of the primary's spectrum and a different measure of significance, tests that accuracy from outside. The public archives also hold more spectra of these stars than the published analyses used: the ESO archive holds 114 HARPS spectra of EBLM J0310−31 above a signal-to-noise ratio of 10, against the 71 of ref. [17], and 264 ESPRESSO spectra of TOI-1338, against 103.

This paper describes such an analysis, the binaries approach of the seti-tools package, which shares its spectra machinery with the package's planet-frame and free-orbit companion searches and their orbitlight release [22], and applies it to public products of the ESO Science Archive alone. Section 2 selects the targets and spectra, Section 3 describes the method, Section 4 compares its results for three systems with published dynamical masses, Section 5 applies it to the southern EBLM binaries with HARPS series, and Section 6 discusses what the comparison shows.

2 Targets and spectra

The targets are the single-lined spectroscopic binaries with archived high-resolution spectra. The Gaia DR3 non-single-star catalogue [23] gives 203 121 single-lined orbits (solution types SB1, SB1C and AstroSpectroSB1) for stars brighter than $G = 13$, and the SB9 catalogue [24] 3 426 single-lined orbits. Each was matched within 10 arcsec to the ESO archive's series of 20 or more spectra of one target from HARPS, ESPRESSO, NIRPS, UVES or FEROS, giving 140 series of 126 stars. A series is searched when its orbit has a period below 100 d covered three or more times, the series holds 30 or more spectra, and the mass function allows a companion of $0.07\,\mathrm{M}_\odot$ or more around a primary of $1\,\mathrm{M}_\odot$. That leaves 68 series. Among them are the HARPS series of 37 EBLM binaries, named here by their EBLM designations, which this paper searches, and TOI-1338's ESPRESSO series; the others belong to O and B stars, interacting binaries with compact companions, and solar-type binaries observed mostly with UVES or FEROS.

The spectra are the archive's Phase 3 one-dimensional products, with barycentric wavelengths and the pipeline's flux and error. For each series, every product with a signal-to-noise ratio above 10 is taken, up to 300 spread evenly over the series' span.

3 Method

3.1 The primary's velocities and orbit

Each spectrum is placed on a grid uniform in velocity at $1\,\mathrm{km\,s^{-1}}$ from 3800 to 7880 Å, divided by its running median over 501 samples, with the telluric bands removed. Its velocity is the peak of its cross-correlation over $\pm 200\,\mathrm{km\,s^{-1}}$ with the line depths of a G0 V template, one minus the median of EBLM J0310−31's 114 HARPS spectra placed at their velocities on its Gaia orbit, refined by a parabola through the three highest samples. The primary's orbit is fitted to these velocities in two steps. A grid over period (within 0.4 per cent of the catalogue's, in steps that move the last epoch by a fiftieth of an orbit), phase (in hundredths), eccentricity (to 0.6) and argument of periastron (in 20° steps) is searched with K_1 and the systemic velocity γ solved linearly at each point, and the best point starts a least-squares fit with a soft loss in period, time of periastron, $\sqrt{e}\cos\omega$, $\sqrt{e}\sin\omega$,

K_1 and γ. The grid makes the fit independent of errors in the catalogue orbit, which for TOI-1338 put the period 0.021 d long and ω 58° away.

3.2 Residuals in the primary's frame

Each spectrum is then moved into the primary's frame at its own measured velocity, placed on the same grid, divided by its running median and by the series' median spectrum, and a cubic in wavelength is removed. A residual shift of a spectrum against the median of $0.5\,\mathrm{km\,s^{-1}}$ or more is taken out before the division. The telluric bands and the sodium sky lines at their telescope wavelengths are masked. Each wavelength's scatter across the series, floored at a tenth of its median, weights the cross-correlation.

3.3 Cross-correlation with an M-dwarf template

The template is an empirical M1 V spectrum: the line depths, one minus the median, of the archived HARPS spectra of GJ 367 [25] in its own frame, negative values set to zero. Each residual is cross-correlated with it by inverse variance over $\pm 300\,\mathrm{km\,s^{-1}}$ at $1\,\mathrm{km\,s^{-1}}$, the result normalised so that a residual of unit noise gives unit scatter. Each cross-correlation row then loses three patterns that are not the companion's: the star's own residual pattern, the median row fitted to each row within $40\,\mathrm{km\,s^{-1}}$ of the star and subtracted; the sky's pattern, the rows' median in the telluric frame fitted back to each row; and five principal components of the rows, which carry the shapes the star's pattern takes from night to night. Each row is weighted by the inverse square of its robust scatter more than $50\,\mathrm{km\,s^{-1}}$ from the star.

3.4 The sum along the companion's track

The companion's barycentric velocity at time t is

$$v_2(t) = \gamma + \Delta v_{\mathrm{sys}} - K_2\,[\cos(\nu + \omega) + e\cos\omega]\,, \tag{1}$$

with ν, ω and e from the primary's fitted orbit and Δv_{sys} the offset of the companion's lines from the primary's systemic velocity. Each row is read at v_2 less the velocity at which its spectrum was placed, by linear interpolation, and the weighted mean of the readings is the trail's value. Readings within $30\,\mathrm{km\,s^{-1}}$ of the primary's lines or within $10\,\mathrm{km\,s^{-1}}$ of the row's telluric velocity are left out, and a value needs readings from at least a fifth of the rows and no fewer than ten. The map of the trail over K_2 from K_1 to $160\,\mathrm{km\,s^{-1}}$ and over Δv_{sys} from -30 to $+30\,\mathrm{km\,s^{-1}}$, both at $0.5\,\mathrm{km\,s^{-1}}$, gives the peak, refined on grids of 0.1 and then $0.05\,\mathrm{km\,s^{-1}}$.

3.5 Significance, uncertainty and masses

The significance is that of the summit of the trail over K_2 at the peak's Δv_{sys}, against the summits of the same curve with the orbit's phase moved by a uniform 0.1 to 0.9 of a period and with whole nights moved to other nights' dates, 300 of each. The lower of the two is quoted, so that what is tied to the orbital phase and what is tied to the hour or the night both stay in a null. The uncertainty of the peak comes from 300 draws of whole nights with replacement, each cleaned again and its peak found on grids of 0.25 and then $0.05\,\mathrm{km\,s^{-1}}$; the 16th, 50th and 84th percentiles are quoted. The masses follow from

$$M_{1,2}\sin^3 i = 1.0361\times10^{-7}\,(1-e^2)^{3/2}\,(K_1+K_2)^2\,K_{2,1}\,P, \tag{2}$$

with the velocities in $\mathrm{km\,s^{-1}}$, the period in days and the masses in $\mathrm{M_\odot}$, divided by $\sin^3 i$ at the published inclination from the eclipses.

Table 1: The primary's orbits fitted to the velocities measured here, against the published orbits. ω is the primary's argument of periastron; rms is the scatter of the velocities about the fit.

System	Source	P (d)	K_1 (km s^{-1})	e	ω (°)	rms (m s^{-1})
TOI-1338	this work	14.608560	21.616	0.1556	117.86	14
	ref. [13]	14.6085579	21.604	0.1555	117.77	
J0310−31	this work	12.642791	27.868	0.3089	185.83	15
	ref. [17]	12.6427926	27.875	0.3087	185.85	
HD 4875	this work	13.635615	29.736	0.1814	251.71	29
	ref. [21]	13.6355611	29.739	0.1811	252.03	

4 Three systems with published dynamical masses

Three of the binaries have published measurements of the secondary's semi-amplitude from other data or other methods: TOI-1338 (EBLM J0608−59), measured from ESPRESSO spectra with Saltire [13]; EBLM J0310−31, measured from HARPS spectra with Saltire [17]; and HD 4875 (EBLM J0050−18), measured from NIRPS spectra by cross-correlation with a synthetic M-dwarf spectrum [21]. The archive holds 264 ESPRESSO spectra of TOI-1338 above the signal-to-noise cut, of which 260 load, 114 HARPS spectra of J0310−31 and 35 of HD 4875.

4.1 Orbits

The orbits fitted to the measured velocities (Table 1) leave a scatter of 14, 15 and 29 m s^{-1}, which is the precision of a cross-correlation velocity against a template built from one star's spectra, and they agree with the published orbits to within 0.012 km s^{-1} in K_1 and 0.0003 in eccentricity. The systemic velocities differ from the published ones by 0.01 to 0.14 km s^{-1}, the offset between the template's zero point and those of the published analyses, which enters the companion's track only through $\Delta v_{\rm sys}$.

4.2 The companions

The companion stands out in each system's map (Fig. 1) at the published K_2 and just redward of the primary's systemic velocity, and the trail over K_2 at that offset rises alone above the nulls (Fig. 2). Table 2 sets the results against the published ones. The semi-amplitudes agree within 0.16 km s^{-1}: 77.27 against 77.16 ± 0.48 km s^{-1} for TOI-1338, 88.12 against 88.27 ± 0.46 for J0310−31, and 75.29 against 75.13 ± 0.16 for HD 4875, whose published value comes from near-infrared spectra alone. At the published inclinations, the masses agree within 0.5 per cent for the primaries and 0.5 per cent for the secondaries. The significances, 19.3, 19.2 and 13.3σ, are each the lower of the two nulls' (Table 2); the null with whole nights shuffled is the stricter one for J0310−31, and the null with the phase moved for TOI-1338 and HD 4875. The profile of the summed rows in the companion's frame (Fig. 3) is a single line 5 to 6 km s^{-1} wide at half maximum, and nothing beyond 15 km s^{-1} of it rises above three times the profile's scatter.

4.3 The companion's systemic offset

The companion's lines sit +1.05, +1.10 and +1.25 km s^{-1} from the primary's systemic velocity in the three systems. The published analyses, which leave the secondary's systemic velocity free and use the pipelines' M2 line mask, found the same sign and size: $+1.40 \pm 0.06$ km s^{-1} in the

Table 2: The secondary's semi-amplitude, the companion's systemic offset, the significance against each null and the masses, against the published values. Ranges are the 16th to 84th percentiles of 300 draws of whole nights; the masses use the published inclinations, 89.639°, 86.531° [17] and 89.057° [21].

	TOI-1338	J0310−31	HD 4875
Spectra (nights), instrument	260 (255), ESPRESSO	114 (114), HARPS	35 (33), HARPS
Span (d)	2231	3031	1376
K_2 ($\mathrm{km\,s^{-1}}$)	77.27 (77.17–77.37)	88.12 (87.97–88.33)	75.29 (74.89–75.49)
Δv_{sys} ($\mathrm{km\,s^{-1}}$)	+1.05 (+1.00 to +1.15)	+1.10 (+1.00 to +1.25)	+1.25 (+0.75 to +1.25)
Significance, phase moved	19.3	20.5	13.3
Significance, nights shuffled	26.2	19.2	14.5
M_1 ($\mathrm{M_\odot}$)	1.102 (1.099–1.106)	1.345 (1.338–1.351)	1.113 (1.102–1.124)
M_2 ($\mathrm{M_\odot}$)	0.3084 (0.3078–0.3090)	0.4252 (0.4238–0.4264)	0.4400 (0.4375–0.4426)
Published K_2 ($\mathrm{km\,s^{-1}}$)	77.16 ± 0.48	88.27 ± 0.46	75.13 ± 0.16
Published M_1 ($\mathrm{M_\odot}$)	1.098 ± 0.017	1.350 ± 0.017	1.1150 ± 0.0066
Published M_2 ($\mathrm{M_\odot}$)	0.307 ± 0.003	0.426 ± 0.003	0.4405 ± 0.0015
Spectra used	103, ESPRESSO	71, HARPS	15 nights, NIRPS
Reference	[13]	[17]	[21]

Table 3: The semi-amplitude and significance from each half of a series by date and from alternate nights, each half cleaned and measured on its own.

Subset	TOI-1338	J0310−31	HD 4875
First half by date	77.17 (15.9σ)	88.02 (8.9σ)	74.24 (5.2σ)
Second half by date	77.52 (14.8σ)	88.32 (14.4σ)	75.99 (10.2σ)
Alternate nights, one set	77.02 (13.1σ)	87.97 (12.0σ)	75.34 (9.3σ)
Alternate nights, other set	77.17 (17.9σ)	87.97 (13.7σ)	75.19 (11.6σ)

ESPRESSO spectra of TOI-1338 [13] and $+1.20 \pm 0.10\,\mathrm{km\,s^{-1}}$ for J0310−31 [17]. The offset collects the difference between the two templates' zero points, the primary's G0 template on the scale of its Gaia orbit and GJ 367's spectrum placed at its pipeline velocities, and the two stars' differences in gravitational redshift and convective blueshift, which for J0310−31's radii [17] amount to about $0.1\,\mathrm{km\,s^{-1}}$ in gravitational redshift. Held at zero instead, the offset moves the summit to 77.62, 88.37 and $74.74\,\mathrm{km\,s^{-1}}$, by +0.35, +0.25 and $-0.55\,\mathrm{km\,s^{-1}}$, since a constant velocity error of the track projects onto K_2 through the phases the spectra sample. At this precision the offset is a parameter of the fit.

4.4 Consistency within each series

Each series divided in two by date, and into alternate nights, gives the semi-amplitudes of Table 3. The halves of TOI-1338 and J0310−31 agree within $0.35\,\mathrm{km\,s^{-1}}$, close to the published uncertainties of 0.48 and $0.46\,\mathrm{km\,s^{-1}}$, which are dominated by splits by observing season [13, 17]. The first half of HD 4875's series, 17 spectra, gives the companion at 5.2σ and $74.24\,\mathrm{km\,s^{-1}}$, and the second half $75.99\,\mathrm{km\,s^{-1}}$; the alternate sets agree within $0.15\,\mathrm{km\,s^{-1}}$.

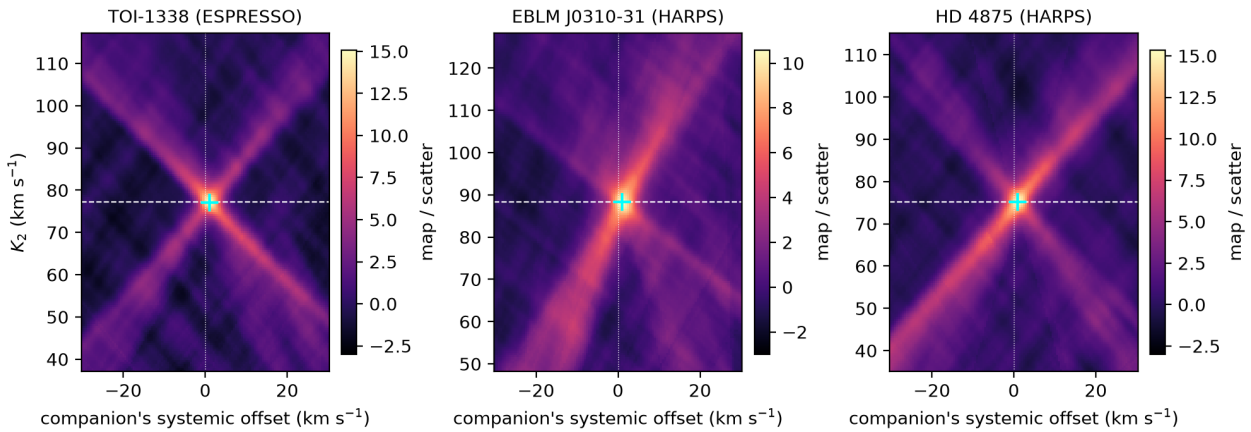


Figure 1: The mean of the cleaned rows along the companion's track over K_2 and the companion's systemic offset from the primary's, in units of the map's scatter away from the peak. The dashed line marks the published K_2, the dotted line zero offset and the cross the peak.

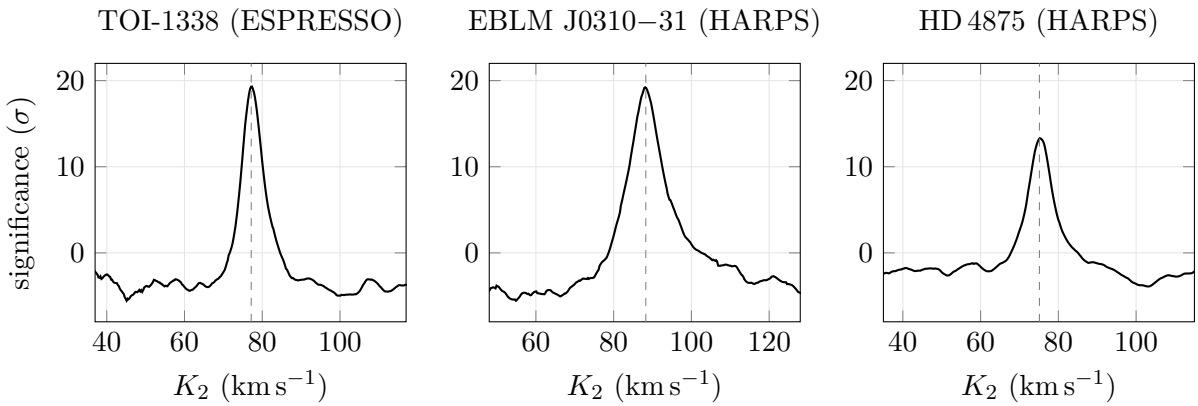


Figure 2: The trail over K_2 at the peak's systemic offset, measured against the summits of the stricter null, so that the summit's height is its significance. The dashed line marks the published K_2.

4.5 Two HARPS series below the threshold

The 73 archived HARPS spectra of TOI-1338 do not show the companion on their own: the trail's summit over K_2 stands 0.0σ above the nulls. Read at the published K_2 and the offset of $+1.0\,\mathrm{km\,s^{-1}}$, they give 2.8σ, with 21 of 2000 null draws as high, as the published analysis of HARPS spectra gave about 3σ [13]. For J1928−38, whose companion of $0.276\,\mathrm{M_\odot}$ was measured at a signal-to-noise ratio of 3.65 in 73 HARPS spectra [17], the 74 archived HARPS spectra give no summit above the nulls and 0.3σ at the published K_2, with 715 of 2000 draws as high. At its brightness, $V = 11.2$, and its companion's mass, this method does not reach J1928−38's secondary in HARPS spectra.

5 The southern EBLM binaries

The same analysis was applied to the 37 EBLM binaries with archived HARPS series, with the companion's systemic offset held at $+1.0\,\mathrm{km\,s^{-1}}$, the value the three systems of Section 4 share, so that each search runs over K_2 alone. Two series cannot be searched: J2126−28 has 17 spectra

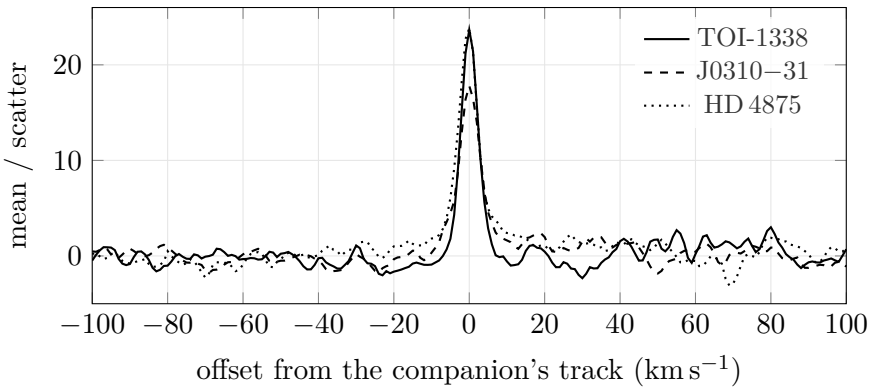


Figure 3: The cleaned rows summed in the companion's frame at the peak's K_2 and systemic offset, in units of the profile's scatter beyond 30 km s^{-1}.

above the signal-to-noise cut and J0543−57 has 21 spectra on six nights. Table 4 lists the other 35. Six stand above 5σ: J0310−31 and HD 4875 of Section 4, and four whose companions have no published semi-amplitude, EBLM J2101−45 at 15.0σ, J0649−27 at 6.7σ, J0825−67 at 6.4σ and J1141−37 at 5.2σ.

Table 5 gives the four measured as in Section 4, with the offset free and the peak bootstrapped over nights. The masses are $m \sin^3 i$; the binaries eclipse, and with the primaries' Gaia DR3 radii the eclipses require $\sin^3 i > 0.993$ for J2101−45 and J0825−67 and $\sin^3 i > 0.96$ for the two with periods near 5 d. J2101−45 is the clearest: each half of its 75 spectra by date, and each set of alternate nights, finds the companion at 8.8 to 15.3σ within 1 km s^{-1} of the full series' K_2, and its companion of 0.556 M$_\odot$ orbits a primary of 1.43 M$_\odot$ whose Gaia radius of 2.67 R$_\odot$ places it above the main sequence. The model-based masses of ref. [4] for J2101−45, 1.29 ± 0.10 and 0.523 ± 0.043 M$_\odot$, J0649−27, 1.45 ± 0.13 and 0.371 ± 0.061 M$_\odot$, and J1141−37, 1.22 ± 0.10 and 0.354 ± 0.050 M$_\odot$, agree with the dynamical ones within their uncertainties. J0825−67 and J1141−37 rest on 27 and 30 spectra, too few to divide, and their orbits fit the measured velocities to 0.15 and 0.13 km s^{-1}, ten times the scatter of the brighter systems. The halves of J0649−27 and J1141−37 by date and by alternate nights each return K_2 within 4.5 km s^{-1} of the full series', at 0.9 to 5.5σ apiece.

Below the threshold, three summits fall at semi-amplitudes that give an F or G primary and an M-dwarf companion: J0415−16 at 4.5σ (1.07 and 0.32 M$_\odot$), J0057−21 at 4.4σ (1.30 and 0.37 M$_\odot$) and J0621−50 at 2.8σ (1.23 and 0.42 M$_\odot$), the last at the K_2 of 109.9 km s^{-1} that its model-based masses predict [4]. The summits of the remaining binaries imply primaries from 0.02 to 5.4 M$_\odot$, as the highest of the noise over a wide range of K_2 does.

6 Discussion

The semi-amplitudes measured here differ from the published ones by −0.15 to +0.16 km s^{-1}, within the published uncertainties, although the analysis makes its own choice at each step. The template is an observed M1 V spectrum rather than a line mask or a synthetic spectrum; the primary's spectrum is removed by division by the series' median in the primary's frame and by principal components of the cross-correlation rows, rather than by a singular value decomposition of the spectra; the significance is measured against nulls built from the same rows rather than from a fit to the map; and the uncertainty comes from resampling whole nights. The agreement indicates that the published dynamical masses of TOI-1338 and EBLM J0310−31 do not depend on those

Table 4: The 35 searchable EBLM binaries with archived HARPS series: the spectra (nights), the span, the primary's orbit fitted to the measured velocities, the summit of the trail over K_2 at a systemic offset of $+1.0\,\mathrm{km\,s^{-1}}$ and its lower significance against the two nulls. Summits of 5σ and more are in bold.

EBLM	Spectra	Span (d)	P (d)	K_1 ($\mathrm{km\,s^{-1}}$)	e	K_2 ($\mathrm{km\,s^{-1}}$)	σ
J0035−69	27 (27)	949	8.4147	17.33	0.244	98.33	0.1
J0050−18	35 (33)	1376	13.6356	29.74	0.181	**75.24**	**13.9**
J0055−00	83 (80)	2771	11.3918	21.16	0.058	118.91	-0.9
J0057−21	35 (35)	2288	32.3014	17.87	0.220	63.37	4.4
J0104−38	35 (35)	1910	8.2561	20.65	0.000	100.40	2.1
J0110−40	43 (43)	1740	28.7816	10.23	0.092	26.98	-1.0
J0310−31	114 (114)	3031	12.6428	27.87	0.309	**88.37**	**19.0**
J0353−16	120 (100)	3960	11.7612	16.64	0.006	83.14	1.0
J0415−16	48 (47)	1390	51.7026	15.21	0.239	50.46	4.5
J0418−53	91 (91)	2909	13.6993	10.36	0.000	109.11	-0.8
J0503−25	40 (39)	1968	15.2652	7.80	0.091	78.80	0.6
J0525−55	61 (61)	1885	4.8001	25.18	0.000	118.18	-0.3
J0526−34	85 (84)	2345	10.1909	23.60	0.126	94.60	2.7
J0608−59	73 (73)	1818	14.6085	21.62	0.156	77.62	-0.1
J0621−50	43 (43)	1778	4.9638	37.58	0.000	109.83	2.8
J0623−27	36 (36)	2493	5.7780	28.05	0.056	190.30	-0.4
J0627−67	41 (41)	1813	9.4689	17.27	0.159	42.27	-1.5
J0649−27	43 (42)	1797	4.3080	32.55	0.002	**119.55**	**6.7**
J0759−69	81 (79)	1988	17.0043	16.26	0.413	23.51	0.1
J0825−67	27 (27)	1234	10.0669	29.43	0.052	**78.43**	**6.4**
J0925−03	61 (61)	2168	14.7748	16.60	0.330	83.10	-0.3
J0946−29	39 (38)	1234	10.4511	20.82	0.161	56.82	-0.9
J0954−45	32 (32)	2262	8.0727	27.83	0.295	112.58	0.2
J1119−39	36 (36)	2257	4.2136	11.13	0.004	29.63	1.6
J1141−37	30 (30)	1718	5.1477	32.23	0.000	**116.23**	**5.2**
J1201−36	65 (65)	2284	9.1131	8.74	0.153	28.99	-0.7
J1219−39	62 (62)	2175	6.7600	10.83	0.056	72.33	-1.2
J1305−31	41 (41)	2123	10.6191	22.40	0.038	82.65	0.1
J1613−76	39 (39)	1308	14.3087	22.90	0.068	82.40	0.4
J1630+10	37 (37)	1872	10.9638	19.44	0.183	84.94	-0.5
J1928−38	74 (73)	2271	23.3230	17.26	0.073	111.76	-1.4
J1959−56	40 (40)	1864	29.7492	17.71	0.365	100.71	-0.9
J2040−41	45 (44)	1901	14.4562	12.45	0.226	31.95	-0.9
J2101−45	75 (75)	2298	25.5768	25.51	0.091	**65.76**	**15.0**
J2207−41	56 (55)	2308	14.7750	8.70	0.069	44.70	0.6

Table 5: The four companions without a published semi-amplitude, measured with the offset free. Ranges are the 16th to 84th percentiles of 300 draws of whole nights; the significances are against the null with the phase moved and the null with whole nights shuffled; the masses are $m \sin^3 i$.

EBLM	Spectra	K_2 ($\mathrm{km\,s^{-1}}$)	Δv_{sys} ($\mathrm{km\,s^{-1}}$)	σ	$m_1 \sin^3 i$ ($\mathrm{M_\odot}$)	$m_2 \sin^3 i$ ($\mathrm{M_\odot}$)
J2101−45	75 (75)	65.41 (65.36–66.01)	+1.45 (+1.10, +1.65)	15.6, 20.6	1.431 (1.412–1.447)	0.556 (0.551–0.559)
J0649−27	43 (42)	119.40 (117.80–121.85)	+1.35 (−0.67, +2.21)	8.4, 6.8	1.230 (1.188–1.296)	0.335 (0.328–0.346)
J0825−67	27 (27)	78.18 (76.98–78.73)	+1.25 (+0.45, +2.05)	7.8, 6.6	0.936 (0.905–0.957)	0.353 (0.346–0.358)
J1141−37	30 (30)	114.68 (113.87–117.18)	+0.15 (−0.40, +1.70)	8.1, 5.5	1.351 (1.296–1.395)	0.376 (0.367–0.384)

analysis choices at the level of their quoted uncertainties, and that the near-infrared semi-amplitude of HD 4875's companion holds in the optical.

The archive's own completeness matters as much as the method. It holds 114 HARPS spectra of J0310−31 above a signal-to-noise ratio of 10, against the 71 of the published analysis, and 260 usable ESPRESSO spectra of TOI-1338, two and a half times the published set. A detection's significance grows with the square root of the number of spectra at fixed quality.

The method has the limits of any optical analysis of these binaries. Toward $0.1\,\mathrm{M}_\odot$ the companion's optical flux falls to 10^{-4} of the primary's and below, and the HARPS series of J1928−38 and of the lowest-mass companions do not reach them. The near-infrared carries a larger share of the companion's light, and the archive holds NIRPS series of nine of the EBLM binaries considered here, taken alongside their HARPS spectra.

7 Conclusion

An analysis of public ESO archive spectra, independent of the published pipelines, recovers the M-dwarf companions of TOI-1338, EBLM J0310−31 and HD 4875 at 19.3, 19.2 and 13.3σ and measures their semi-amplitudes as 77.27, 88.12 and $75.29\,\mathrm{km\,s^{-1}}$, within $0.16\,\mathrm{km\,s^{-1}}$ of the published values, with masses within 0.5 per cent of the published dynamical masses. HD 4875's companion, measured before in the near-infrared alone, is detected here in the optical. The companion's lines sit +1.05 to $+1.25\,\mathrm{km\,s^{-1}}$ from the primary's systemic velocity in all three, and leaving that offset free is needed for accuracy at the tenth of a kilometre per second. Across the 35 searchable southern EBLM binaries with archived HARPS series, the companions of EBLM J2101−45, J0649−27, J0825−67 and J1141−37 are measured for the first time, J2101−45's at 15.0σ with minimum masses of 1.431 and $0.556\,\mathrm{M}_\odot$.

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Data availability

The spectra are public in the ESO Science Archive. The analysis is the binaries approach of the seti-tools package that holds this paper (`seti.py composition binaries`), whose coverage store records the run as `binaries_eblm_known_orbits`; the measurements, the maps and the survey's results are in the paper's data folder.

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